

Morteza Maghrebi

Automotive Embedded Systems & R&D Engineer

🔧 Embedded Systems

📱 Android Development

📋 Project Management

🚗 Automotive Electronics

📍 Tehran, Iran | Open to EU Relocation

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🏠 Overview

📁 Experience

🔖 Projects

🔗 Skills

👤 Professional Summary

Automotive R&D and Embedded Systems Engineer with 5+ years of experience in digital instrument clusters, CAN bus systems, bootloaders, Linux-based boards, and industrial hardware development. Proven ability to deliver end-to-end solutions from low-level microcontroller programming to production test systems. Technical lead and project manager for key automotive projects.

🎓 Education

MSc in Nuclear Engineering

Medical Radiation Specialization
Amirkabir University of Technology
2016 – 2019

BSc in Electrical Engineering

Control Systems Specialization
Shahrood University of Technology
2011 – 2015

5+

Years Experience

30+

Projects

100%

Success Rate

📁 Key Experience

Arg Telecom

Automotive R&D Engineer
2022 – Present

Azar Hesgar Novin Co.

Electronics Engineer & Software Developer
2019 – 2022

Freelancer

Android Developer
2020 – 2021

🔗 Core Skills

STM32

CAN Bus

C/C++

Android

C#

Linux Embedded

🔖 Featured Projects

Digital Instrument Cluster – Dena Vehicle

Lead Engineer

Full development of automotive digital instrument cluster including low-level programming, HAL, bootloader, CAN-based update system, high-level...

STM32 CAN C++ Qt

Dual-Bus CAN Analyzer + C# Application

Full Development

Designed a dual MS-CAN/HS-CAN analyzer with reliable high-load communication and real-time on-device diagnostics.

C++ C# Altium Designer

VANET Vehicle Network Simulation

C++ Simulation

Simulated vehicular ad-hoc network where packets are passed between vehicles to reach destination using 802.11p and GPS.

C++ Networking Simulation

Sensor Calibration Production Line

2019-2021

Developed hardware and software for pressure sensor calibration production line with full automation and quality control.

C# Embedded Automation

Detailed Portfolio

📁 Professional Experience

Arg Telecom

Automotive R&D Engineer

2022 – Present

- Automotive research and development focused on digital instrument clusters and embedded automotive electronics.
- Designed and implemented firmware, bootloaders, CAN-based update systems, and production test equipment.
- Led full software lifecycle from low-level MCU programming to PC and Linux-based user interfaces.
- Established and administered the company GitLab infrastructure, serving as system administrator.

Azar Hesgar Novin Co.

Electronics Engineer & Software Developer

2019 – 2022

- Designed industrial embedded devices for pressure control, sensor calibration, and leak testing.
- Developed C# desktop software and DLLs for hardware communication.
- Implemented custom communication protocols and reverse engineering solutions.

Freelancer

Android Developer

2020 – 2021

- Developed network-based Android applications using TCP/UDP sockets.
- Implemented multithreading and performance-optimized communication systems.

Digital Instrument Cluster – Dena Vehicle

Lead Embedded Engineer & Project Manager

STM32 CAN C Qt Linux

- Programmed low-level hardware layer for a new automotive microcontroller using direct register-level coding
- Developed custom HAL (Hardware Abstraction Layer) for the MCU
- Implemented CAN-based firmware update via diagnostic tools
- Designed and developed cluster bootloader, Windows PC update software, standalone SD-card-based MCU updater
- Led Linux-based UI development using Qt and designed UI graphics using GIMP

VANET Vehicle Network Simulation

C++ Network Simulation | 2021-2022

Sensor Calibration Production Line

Pressure Sensor Calibration System | 2019-2021

Silo Temperature & Humidity Monitoring

Team Project | 2020-2022

Smart Parking Door Control

Android & Embedded System | 2021-2022

ECG Signal Amplifier for Cardiology

Analog Circuit Design | 2021

Dena Cluster Tester (Production & Vehicle Test Stand)

Automotive Test System

CAN Bus Analyzer (MS-CAN & HS-CAN)

CAN Diagnostic Tool

Linux Board Programmer – i.MX6UL

Key Vehicle Project

Delta 3D Printer

Mechanical & Embedded Design

Laser Engraving & Cutting Machine

Industrial Machine Design

Word Hero – Language Learning Super App

Mobile & Web Application

Sleep Hypnosis & Meditation App

Audio Wellness Application

 Technical Skills

STM32

CAN Bus

C/C++

Android (Java)

C#

Qt

Linux Embedded

Bootloaders

UART/SPI/I2C

Git/GitLab

Python

SolidWorks